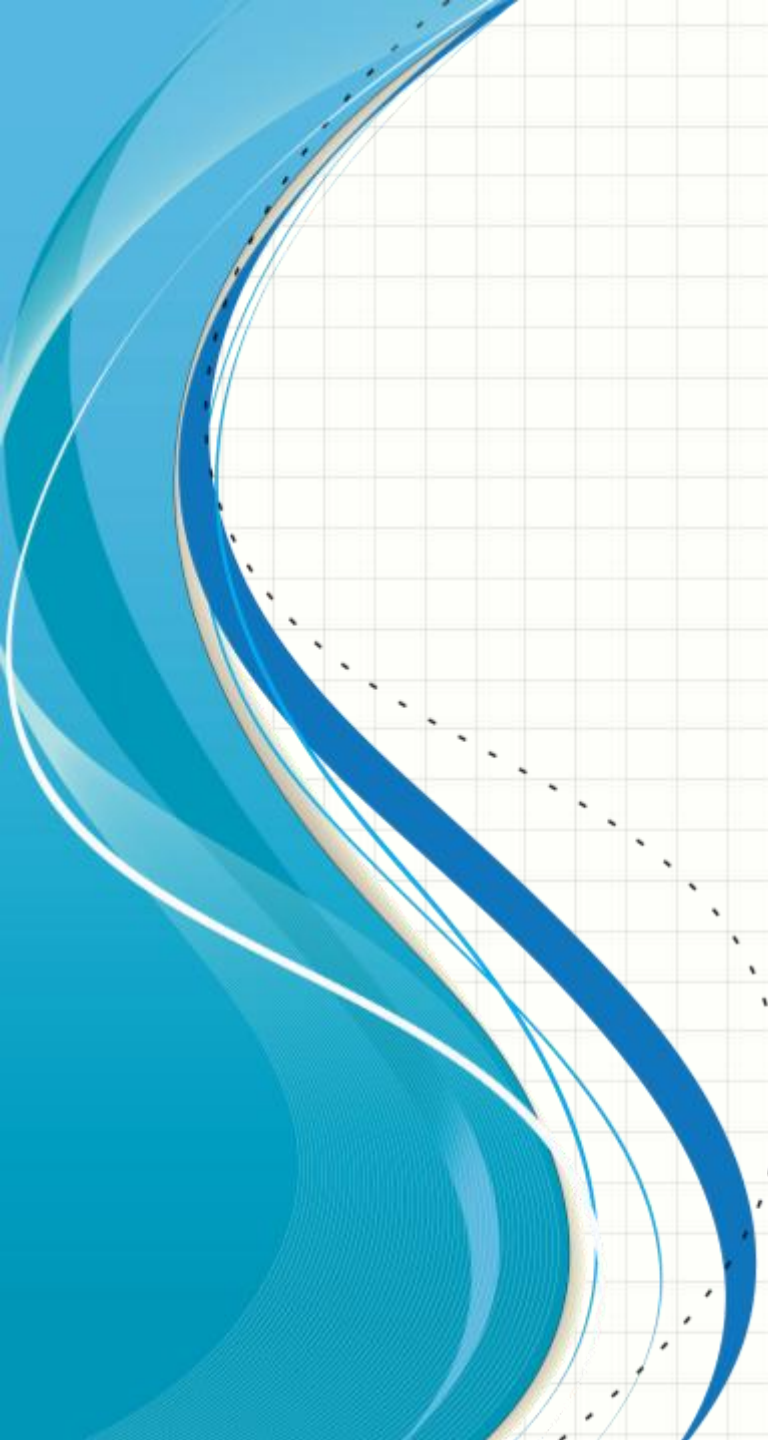




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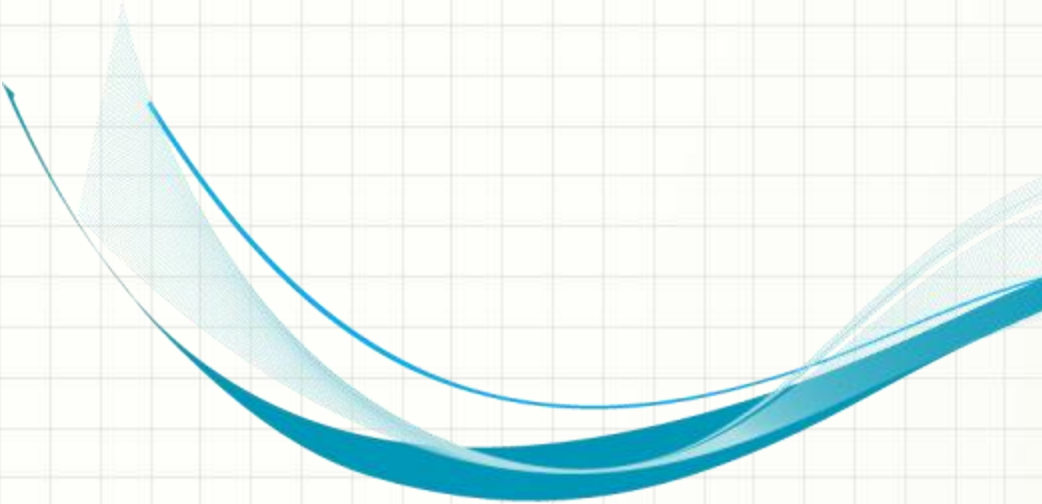
SOFTWARE ENGINEERING

CS 487

Prof. Dennis Hood
Computer Science

Lesson Overview

- User-Centered Design
- Reading
 - Ch. 23 – Project Planning
- Objectives
 - Explore the concept of “user experience” and software engineering’s role in creating it
 - Explore the user-centered design process
 - Discuss UCD as a means of achieving measurably better HCI
 - Examine the relationship between UCD and situational awareness (SA)



Week 14

User-Centered Design

HCI and UCD

- Human-computer interface
 - The means of interaction for human users
 - Automation would in effect repurpose the HCI as a computer-computer interface
- User-centered design
 - A design methodology which both
 - involves the user in the design process and
 - emphasizes user performance in the evaluation of design

Give the People What They Want?

- The user's opinion is more important than yours
- Appreciate users' capabilities
- Offer help – in many forms
- Strive for quality user experiences
- Involve the user early
- Use what works
- Things can be interpreted differently by different people at different times

The Design Process

- Identify user groups
- Establish requirements for the user experience
- Develop alternative designs
- Create interactive prototypes
- Evaluate with relevant, objective analysis
- Collect feedback and fix it in the next version

Responsibilities of the Designer

- Users of your product should enjoy the experience
- The product should undeniably ease / improve the user's life
 - Increased productivity
 - Reduced anxiety / minimal frustration
 - Better performance / results
 - Consistency, integrity, reliability, etc.
- The product should exceed all competing products in these areas
 - Customers will be willing to pay more
 - Customers will buy variant products
 - Customers will tell their friends good things

Usability Goals

- Effective to use
- Efficient to use
- Safe to use
- Having good utility
- Easy to learn
- Easy to remember how to use

Design Principles

- Make it visible, or not, as needed
- Provide feedback
- Set boundaries (constraints)
- Be consistent
- Provide clues

Evaluation Described

- Evaluation is the process of assessing the goodness/acceptability of a design
- Why
 - Evaluation is critical to achieving user acceptance
 - It is beneficial to get it right the first time
- What
 - Criteria should match the users' needs/interests
 - Easy to learn, fast, satisfying, entertaining, etc.
- Where
 - My place or yours? Control vs. comfort
- When
 - You can evaluate at any point, but should you?
 - Does it meet standards? Does it meet user's needs?

Evaluation Approaches

- Usability testing
 - Quantification of user performance
 - Time to complete, error rates, type/severity of errors, etc.
 - Measure *typical* users' performance on *typical* tasks
 - Controlled by the evaluator
- Field studies
 - See how users act and interact (with each other, a given product, etc.) in their *natural* setting
- Analytical evaluation
 - Heuristic (guidelines and standards) evaluations
 - Walkthroughs of scenarios using prototypes
- Hybrids

Evaluation Methods

- Observation and inquisition
 - Observe users
 - Ask users, ask experts
 - Observation, questionnaires, interviews
- User testing
 - E.g., based on scenarios
- Inspections
 - E.g., based on heuristics
- Modeling
 - To predict and establish benchmarks

Usability Testing

- Testing the product
 - to determine the extent to which it is usable
 - by the intended user population
 - on the tasks for which it was designed
- User testing
 - Measures human performance on specific tasks
 - Logging of keystrokes and mouse movements, video recordings, etc.
- User satisfaction questionnaires / interviews
 - How do you feel?
 - Efficiency and effectiveness

Measuring Usability

- Time
 - Time to complete a defined task
 - After a specified time away from the product
- Number
 - Number and type of errors made per task
 - Number of errors per unit time
 - Number of navigations to help
 - Number of users making a particular error
 - Number of users completing successfully

User-Centered Design

- Decision support
 - Traditionally technology-centered
 - Sensors, reports, gauges, alarms, etc.
- vs. Information overload
 - Bottlenecks
 - 7 +/- 2 chunks
 - Recall time, processing time
- Operator error
 - A causal factor of 60% to 85% of all accidents
- UCD is improving performance and acceptance

UCD and Situation Awareness

- UCD is not
 - Giving users everything they ask for
 - Making decisions for them
 - Doing things for them
- Design principles
 - Organize technology around the user's needs and capabilities
 - And around the way they process information and make decisions
 - Keep the user in control and aware of state
 - thereby reducing anxiety and
 - improving decision-making effectiveness

Design Principles

- Provide support for projection
- Provide support for confirmation
- Remove ambiguity
- Reduce false alarms
- Set trade-offs appropriately
- Use multiple modalities, consistently
- Minimize disruptions
- Support assessment and diagnosis of multiple alarms
- Support global SA

Keep It Simple

- Again, working memory is limited – designing with the limitations in mind is critical
- Less is more (slick, cool, flashy are not)
 - Avoid clutter
 - Avoid unnecessary marks, elements, etc.
 - Avoid scrolling, paging, etc.
 - Every bit of ink (pixel) requires a reason
- Organization is key
 - Group logically and summarize where possible
 - Focus on context
 - Rely on iconic memory where possible